

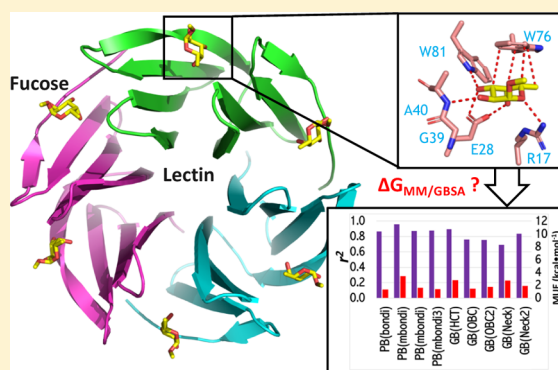
Assessing the Performance of MM/PBSA, MM/GBSA, and QM–MM/GBSA Approaches on Protein/Carbohydrate Complexes: Effect of Implicit Solvent Models, QM Methods, and Entropic Contributions

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Supporting Information

ABSTRACT: Rapid and accurate binding affinity prediction of protein–carbohydrate complexes is a major challenge in glycomimetics design. Among the existing computational techniques, end-point methods have received considerable interest because of their low computational cost. However, significant obstacles remain when such methods are applied to protein–glycan complexes. This article reports the performance of end-point free-energy calculation methods: molecular mechanics/Poisson–Boltzmann surface area (MM/PBSA), MM/generalized Born surface area (MM/GBSA), and quantum mechanics–MM/GBSA (QM–MM/GBSA) on monosaccharides bound to RSL lectin from *Ralstonia solanacearum*. A careful investigation of the molecular dynamics simulation length, van der Waals radii sets, GB models, QM Hamiltonians, and entropic compensation has been made, and the results are compared with the experimental binding free energies from isothermal titration calorimetry/surface plasmon resonance measurements. The binding free energies using implicit solvent methods are found to be sensitive to the simulation length, radii set, GB model, and QM Hamiltonian. A simulation length of 10 ns using the radii set *mbondi* provides the best agreement with the experimental values ($r^2 = 0.96$) by MM/PBSA. The GB^{HCT} model is in accord with the experimental values in MM/GBSA ($r^2 = 0.91$) or in combination with parameterized model number 6 (PM6) ($r^2 = 0.98$) in QM–MM/GBSA. Out of 12 QM Hamiltonians tested, PM6, density functional theory-based tight binding (DFTB), and their variants proved to be more efficient than other semiempirical methods. These methods perform equally well in predicting both absolute and relative binding free energies.



1. INTRODUCTION

Glycans are one of the highly abundant biomolecules contributing significantly to the dynamics, structure, and function of the proteins. For example, cell surface glycans serve as a marker for cell–cell, cell–matrix, and host–pathogen interactions.¹ Most of these processes occur because of the specific interaction of glycans with biomolecules including enzymes, antibodies, and other proteins. The interactions between glycans and glycan-binding proteins called lectins are known to mediate host–pathogen recognition.² Lectins have high specificity for particular monosaccharide moieties, and they can often distinguish minuscule difference among the monosaccharides. Regardless of the proven role of glycans in the pathophysiology of major diseases,^{3,4} the development in this field is still lagging behind. One of the main deterrents in the glycan research is the structural complexity of glycans. Understanding how glycans are recognized by lectins is of the fundamental issues in glycosciences. This can be achieved by getting insights into the structure, dynamics, and function of glycans and their noncovalent interaction with proteins at the atomic level.

Therefore, development of methods to explore protein–glycan interactions at the molecular level is essential.

Computational modeling and binding free-energy prediction are currently receiving considerable interest to address the fundamentals of molecular recognition.^{5–7} However, because of several unusual structural features of glycans (i.e., anomerization, flexibility, and a high number of hydroxyl groups), lectins (multiple ligand binding sites), and lectin–glycan complexes (stacking interactions), it is difficult to predict the binding free energies of protein–glycan complexes accurately.^{8,9} Despite recent developments in protein–glycan binding free-energy prediction methods, ranging from end-point approaches such as linear interaction energy (LIE),¹⁰ molecular mechanics/Poisson–Boltzmann surface area (MM/PBSA), and MM/generalized Born surface area (MM/GBSA)^{7,9,11} to rigorous approaches such as alchemical free energy methods¹² or potential of mean force-based approaches,¹³ their application has been limited. In a recent

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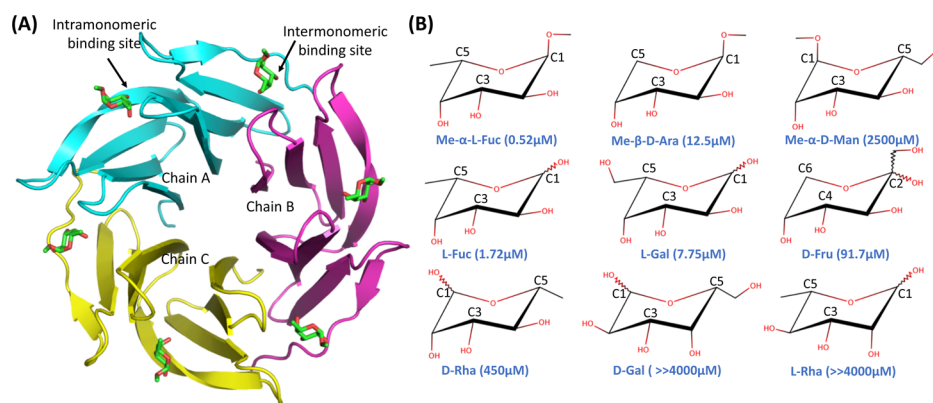


Figure 1. Trimeric structure of RSL lectin (A) and schematic representation of the monosaccharides used in the binding free-energy calculations (B). The K_d value of the monosaccharide binding to RSL is provided in brackets. Both α - and β -anomers have been used for the calculations if possible.

study, the thermodynamic integration (TI) method performed quite well on lectin–glycan complexes, but the accuracy was limited to smaller appearing/disappearing glycan functional groups (e.g., hydroxyls).¹² For slightly larger functional groups such as hydroxyethyls, perturbation had to be performed in two steps. Despite getting quite accurate predictions for monosaccharides, TI currently appears to be a less suitable choice to handle glycans where appearing/disappearing groups can be even full monosaccharide units. The MM/PBSA and MM/GBSA methods are an attractive alternative owing to their pairwise nature that allows decomposition of the free energy components and calculation of per-residue contribution to the binding free energy. This makes the MM/PBSA and MM/GBSA methods a suitable choice for modeling protein–glycan interactions as the binding free-energy contribution of each monosaccharide unit can be easily estimated.¹¹ The MM/PBSA and MM/GBSA methods have been successful in many applications, but both often struggle to capture small differences in binding energies.

Regardless of successes in binding free-energy calculations of nonglycan ligands by MM/PBSA and MM/GBSA, their application to other classes of biomolecular interactions, such as lectin–carbohydrate complexes, has been less explored.¹⁴ The performance of MM/GBSA on protein–glycan complexes often shows poor correlation between the experimental and predicted free energies, and numerous outliers can be found.^{7,15} The precision of this kind of methods is mostly hampered by the high fluctuations in the calculation of entropic contribution. However, recent work has shown that this variance can be reduced to a third, which encourages us to step forward and explore the applicability of those methods to protein–glycan complexes.¹⁶ Thus, we aim to explore if a particular implicit solvent model, or a combination of other parameters including entropic calculation approaches, offers a better and more reliable agreement with the experimental free energies in lectin–glycan complexes. The ultimate goal for such methods is to be able to predict the binding free energies of the di-, tri-, or oligosaccharides to branched N-glycans. To avoid the sampling errors and approach the problem systematically in steps, we choose only monosaccharides to explore the (I) performance of several GB models in MM/GBSA, (II) effect of atomic radii sets in MM/PBSA calculations, (III) performance of semiempirical quantum mechanics (QM) approaches in QM–MM/GBSA, and (IV) overall role of entropies in reproducing experimental binding

free energies by MM/PBSA, MM/GBSA, and QM–MM/GBSA in protein–monosaccharide complexes.

2. THEORY AND METHODS

2.1. Preparation of Molecular Models. **2.1.1. Complex Preparation.** Lectin RSL from *Ralstonia solanacearum* is a six-bladed β -propeller trimeric lectin, with 90 amino acid residues in each monomeric chain. The initial coordinates of the RSL bound to Me- α -L-fucose (MeFuc) were obtained from the X-ray crystal structure (PDB ID: 2BT9).¹⁷ Each RSL monomer unit contains one L-fucose binding site located in between the two β -sheet blades called the intramonomeric binding site, and the other is formed at the interface between the neighboring monomers called the intermonomeric binding site (Figure 1a). In total, there are six symmetrically arranged binding sites reported in the crystal structure. Isothermal titration calorimetry (ITC) has suggested that the intermonomeric and intramonomeric binding sites are indistinguishable.¹⁷ Thus, we model monosaccharides in the intramonomeric binding site only. To check the binding diversity in the RSL lectin, we have calculated ΔH of MeFuc in each binding site in the presence and absence of ligands in the remaining binding sites.

A data set of monosaccharides, spanning from binders, low-affinity binders to nonbinders, was selected (Figure 1b). The experimental binding constants (K_a/K_d) of these monosaccharides have been previously measured using ITC/surface plasmon resonance (SPR) experiments.^{17,18} The three-dimensional coordinates of the ligands were modeled using the GLYCAM Carbohydrate Builder web server.¹⁹ The starting structures of the RSL–saccharide complexes for other monosaccharides were prepared manually by superposing their ring atoms (C1, C2, C3, C4, C5, and O5) to MeFuc while keeping the orientation of O2, O3, and O4 hydroxyls unchanged where possible. The monosaccharides with free hydroxyl at the anomeric position in ITC/SPR experiments were modeled in their α - and β -form. The starting conformation of the monosaccharide was kept in its most favorable chair form because the potential energy barrier for ring flips in pyranoses can be ca. 3–5 kcal·mol^{−1} or greater.²⁰

2.1.2. Molecular Dynamics Simulations. The input files for molecular dynamics (MD) simulations were prepared using AmberTools14.²¹ The protein was described with an Amber ff14SB force field and glycans by a GLYCAM06 force field (version j-1).²² All systems were solvated in a rectangular box

of TIP3P water molecules extending 11 Å away from the edges of the solute using a program solvate.²³ The total charge of each system was set to zero, and isotonic conditions were maintained by adding ions (Na^+ and Cl^-) by the Debye–Hückel rule in the solvate. The simulation systems were equilibrated using multistep protocol published before.¹¹ Finally, a 10 ns NPT production run at 300 K was performed with the following MD settings: temperature scaling by Langevin dynamics (collision frequency = 2), pressure relaxation every 1.2 ps, SHAKE constraints, nonbonded interaction cutoff of 9 Å, and 2 fs integration time step. The sander and *pmemd.cuda* programs of Amber14 were used to perform the equilibration and production run, respectively. Atomic coordinates were saved every 10 ps and were subsequently used for MM/PBSA, MM/GBSA, and QM–MM/GBSA calculations.

2.1.3. Binding Free-Energy Calculation. Binding free energies were calculated using the MMPBSA.py module of AmberTools14. A total 100 frames taken at a regular time interval from the MD simulation trajectory were used for the binding free-energy calculations in each case.

The binding free energies were calculated using the following equation:

$$\Delta G_{\text{bind}} = \Delta H - T\Delta S \approx \Delta E_{\text{MM}} + \Delta G_{\text{solv}} - T\Delta S$$

The MM term ΔE_{MM} accounts for the bonded (ΔE_{bond} , ΔE_{angle} , and $\Delta E_{\text{dihedral}}$) and nonbonded (ΔE_{ele} and ΔE_{vdW}) energy terms. The solvation energy term (ΔG_{solv}) is calculated as the sum of the polar contribution to solvation free energy calculated by the GB model ($\Delta G_{\text{PB/GB}}$) and the nonpolar contribution calculated from a linear relation to the solvent-accessible surface area (ΔG_{SA}). The $\Delta G_{\text{PB/GB}}$ is calculated using PB (in MM/PBSA) or GB (in MM/GBSA) equations, whereas, in the QM–MM/GBSA approach, ΔG_{bind} is calculated as

$$\Delta G_{\text{bind}} \approx \Delta E_{\text{MM}} + \Delta E_{\text{SCF}} + \Delta G_{\text{solv}} - T\Delta S$$

where ΔE_{SCF} encompasses the energy of atoms in the QM zone calculated using various semiempirical QM Hamiltonians. The overall agreement between the experimental and predicted values was assessed by the comparison of Pearson's correlation coefficient,

$$\left(\sum (x - \bar{x})(y - \bar{y}) / \sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2} \right), \text{ and mean}$$

unsigned error (MUE) ($(\sum |\Delta G_{\text{exp}} - \Delta G_{\text{cal}}|) / n$) for the full data set as proposed by Pearlman and Charifson.²⁴ Non-methylated monosaccharides were modeled in both α - and β -anomeric configuration, and the one that resulted in more favorable binding free energy to protein was assumed to be the one binding to RSL. The absolute binding free energy of binding anomers was only used for the calculation of statistical parameters r^2 and MUE. Additionally, we also calculated the change in the binding energies ($\Delta \Delta G_{\text{cal}}$) of monosaccharides relative to MeFuc. The performance of each method was calculated in terms of their accuracy to predict both absolute and relative binding free energies (ΔG_{cal} and $\Delta \Delta G_{\text{cal}}$). Other specific parameters used in each method are described below in particular sections.

2.1.3.1. MM/PBSA Calculations. The MM/PBSA binding free energies were calculated using bondi, mbondi, mbondi2, and mbondi3 radii sets one by one. The surface tension and nonpolar solvation free-energy correction terms were set to 0.00378 kcal·mol^{−1} and 0.5692, respectively. An offset value of

−0.05692 was used to correct the nonpolar solvation free-energy contribution. The interior and exterior dielectric constants were set to 1 and 80, respectively. The ratio between the longest dimension of the rectangular finite difference grid and that of the solute was set to 4.0. An ionic strength of 0.15 mM was used in the PB equation. The solvent probe radius and grid spacing were set to 1.2 and 2, respectively. The remaining parameters were set to their default values, as stated in the Amber14 manual.²¹

2.1.3.2. MM/GBSA Calculations. We investigated five GB models commonly referred to as GB^{HCT},²⁵ GB^{OBC} and GB^{OBC2},²⁶ GB^{NECK},²⁷ and GB^{NECK2}.²⁸ The radii set for various GB models was used according to the recommendation in the AMBER manual: *mbondi* in GB^{HCT} (igb = 1), *mbondi2* in GB^{OBC} (igb = 2) and GB^{OBC2} (igb = 5), *bondi* in GB^{NECK} (igb = 7), and *mbondi3* in GB^{NECK2} (igb = 8). The salt concentration was set to 0.15 M. The surface tension and nonpolar solvation free-energy correction terms were set to 0.005 kcal·mol^{−1} and 0.0, respectively. An offset value of 0.0 was used to correct the nonpolar solvation free-energy contribution. The interior and exterior dielectric constants were set to 1 and 80, respectively. Other parameters were set to their default values in Amber14.

2.1.3.3. QM–MM/GBSA Calculations. The monosaccharide and seven neighboring amino acid residues (Arg17, Gly28, Trp76, Trp81, Ile61, Gly39, and Ala40) were assigned to the QM zone. The total charge of the QM zone was set to zero. The QM zone was treated one by one using Austin model number 1 (AM1),²⁹ AM1-DH+,³⁰ modified neglect of diatomic overlap (MNDO),³¹ MNDO/d, parameterized model number 3 (PM3), PM3-CARB,³² parameterized model number 6 (PM6),³³ PM6/d, PM6-DH+, reparameterization of AM1 (RM1),³⁴ density functional theory-based tight binding (DFTB), and self-consistent charge (SCC)-DFTB³⁵ Hamiltonians. The QM/MM system was minimized using a self-consistent field convergence criterion of 0.1×10^{-7} kcal·mol^{−1}. The electrostatic interaction between the QM and MM zones was coupled using an electrostatic embedding scheme (*qmmm_int* = 1). The performance of each QM approach is evaluated in combination with all the five GB models discussed above in the MM/GBSA Calculations section. The dispersion parameters for H, C, N, O, P, and S for the DFTB and SCC version (SCC-DFTB) Hamiltonian were taken from the DFTB web page.³⁶ Other parameters for the GB part of the QM–MM/GBSA calculations were kept the same as in MM/GBSA calculations.

2.1.3.4. Entropy Calculations. The entropy calculations were performed using a normal mode (NM) analysis at the MM level.²¹ A total of 96 frames from the MD trajectory of 10 ns were used under the following setting: *mbondi2* radii, and the dielectric-dependent constant was set to 1. The energy gradient for the convergence criterion of the minimization was 0.001 with maximum 10 000 cycles. We calculated the entropies using the harmonic approximation (NMs) approach (S_{QHA}) of AmberTools14. In addition to this, the entropy using quasi-harmonic approximation approach (S_{QHA}) was also calculated by *cptraj*.

3. RESULTS AND DISCUSSION

The MD trajectories of the RSL–monosaccharide complexes were analyzed, and the root-mean-square deviation (RMSD) of nonhydrogen atoms indicated that the system is equilibrated well and stable (Figure 2). The experimental ΔG_{exp} and the

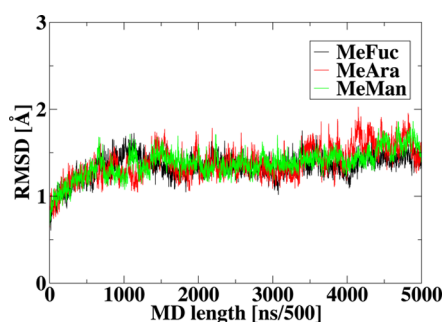


Figure 2. RMSD of the nonhydrogen atoms of RSL in complex with Me- α -L-Fuc (MeFuc), Me- β -D-Ara (MeAra), and Me- α -D-Man (MeMan).

calculated enthalpic and entropic contributions to binding energies (ΔH and $T\Delta S$) from MM/PBSA and MM/GBSA are listed in Table 1. As considered before,¹² the binding free energy of nonbinders (D-Gal and L-Rha) is assumed to be -2.1 kcal·mol⁻¹ for the comparison of r^2 and MUE with the previous study. Because MM/PBSA- and MM/GBSA-type calculations are better suited for ligand ranking, the entropic contributions to the binding free energy are often neglected. Thus, the r^2 and MUE between ΔH and ΔG_{exp} have also been shown.

The ΔH of MeFuc is slightly different in each binding site, but these differences can be safely ignored as the differences are within the range of the standard deviation of those values. Additionally, the binding free energy of MeFuc is unaffected by the presence or absence of the ligands in other binding sites, indicating no binding cooperativity in RSL (Table S1).

3.1. Efficacy of MD Simulation Length. The r^2 between the experimental and predicted ΔH at different MD trajectory lengths for various GB models and PB models with different radii sets is shown in Figure 3. Both MM/PBSA and MM/GBSA show improvement in r^2 for MD trajectory up to 7 ns, but only marginal changes are observed between 8 and 10 ns. This suggests that a minimum of 7 ns MD simulation is

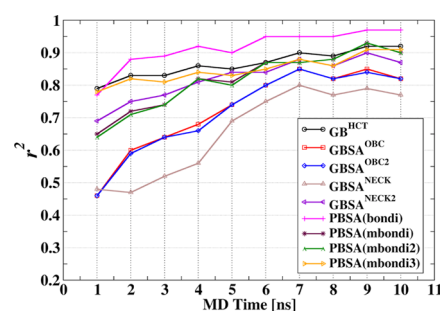


Figure 3. Plot of the coefficient of determination (r^2) between the computed enthalpy and experimental free energies for different lengths of MD simulation. The number of frames (100) used to calculate the enthalpy is the same for every calculated data point.

required to obtain converged energies. Interestingly, there is no quick downfall in r^2 for longer MD trajectories as observed by Su et al. 2015 for ligands treated with generalized AMBER force field (GAFF).³⁷ This decrease can be attributed to a low accuracy in the prediction of ligand binding conformation. A weaker agreement with the experimental data for longer simulation trajectories for general ligands can be expected if the force field parameters are not appropriate to reflect a realistic protein–ligand-bound conformation during MD simulations.

The GAFF parameters may not be always of high quality for all the ligands. In longer simulations, it may lead to structural deformations and ultimately to a weaker agreement with the experimental binding energies. Instead, numerous short simulations help to mitigate the force field inadequacy and result in a better agreement. Thus, these results suggest that GLYCAM06 parameters are of high quality as longer simulations provide better conformational sampling, along with improved r^2 and converged energies.

3.2. Effect of Radii on MM-PBSA. In general, MM/PBSA calculations have low sensitivity to radii sets (Table 2). The differences shown for protein–glycan complexes are also not

Table 1. Experimental (ΔG_{exp}) and Calculated Binding Free Energies (ΔH_{cal} and $T\Delta S_{\text{cal}}$ Contribution to Binding Free Energies) for Monosaccharides Calculated Using GB^{HCT}, GB^{OBC}, GB^{OBC2}, GB^{NECK}, and GB^{NECK2} Models in MM/GBSA and Various Radii Sets (mbondi, mbondi2, bondi, and mbondi3) in the MM/PBSA Approach^a

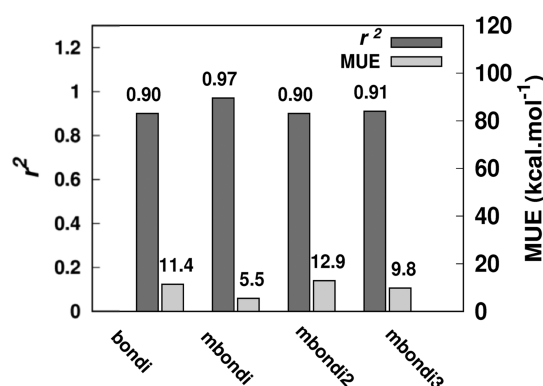
ligand	ΔG_{exp}	ΔH_{cal} (MM/GBSA)					ΔH_{cal} (MM/PBSA)				$T\Delta S_{\text{cal}}$	
		GB ^{HCT}	GB ^{OBC}	GB ^{OBC2}	GB ^{NECK}	GB ^{NECK2}	mbondi	mbondi2	bondi	mbondi3	$T\Delta S_{\text{HA}}$	$T\Delta S_{\text{QHA}}$
Me- α -L-Fuc	-8.6	-32.3	-37.6	-41.8	-41.7	-31.6	-17.5	-23.1	-21.4	-19.8	-20.6	-12.3
α -L-Fuc	-7.9	-28.9	-35.1	-39.2	-38.7	-28.7	-15.6	-21.6	-20.0	-18.3	-16.8	-12.3
β -L-Fuc	-7.9	-28.3	-33.2	-36.7	-33.5	-27.3	-14.0	-21.0	-19.3	-17.8	-17.6	-12.3
α -L-Gal	-7.0	-30.2	-36.2	-40.2	-38.9	-29.4	-15.0	-21.6	-19.9	-18.2	-16.7	-12.4
β -L-Gal	-7.0	-27.7	-34.4	-38.5	-35.7	-27.1	-12.1	-20.6	-19.0	-17.4	-19.1	-12.4
Me- β -D-Ara	-6.7	-27.1	-32.8	-36.8	-35.4	-26.7	-12.2	-18.1	-16.5	-14.9	-17.4	-12.3
α -D-Fru	-5.5	-25.1	-32.0	-36.2	-34.1	-24.8	-8.4	-17.3	-15.8	-14.1	-18.9	-12.4
β -D-Fru	-5.5	-25.9	-33.5	-37.6	-37.2	-27.1	-11.0	-19.2	-17.7	-15.9	-19.4	-12.4
α -D-Rha	-4.6	-21.5	-28.1	-31.1	-29.7	-21.9	-7.9	-15.2	-13.9	-12.2	-16.7	-12.3
β -D-Rha	-4.6	-24.9	-30.1	-33.6	-31.4	-24.0	-9.8	-16.5	-15.0	-13.3	-16.5	-12.3
Me- β -D-Man	-3.5	-25.1	-32.5	-36.3	-35.5	-25.8	-7.6	-17.1	-15.6	-13.8	-17.6	-12.4
α -D-Gal	-2.1	-20.4	-28.9	-32.6	-31.6	-22.6	-2.2	-11.9	-10.7	-8.9	-18.2	-12.4
β -D-Gal	-2.1	-18.1	-26.1	-29.6	-30.8	-21.2	-3.2	-13.0	-12.0	-10.5	-16.8	-12.4
α -L-Rha	-2.1	-17.8	-24.8	-27.7	-27.2	-19.0	-4.5	-12.6	-11.6	-10.0	-16.0	-12.3
β -L-Rha	-2.1	-21.2	-29.5	-33.1	-31.3	-21.8	-5.6	-14.3	-13.0	-11.1	-18.6	-12.3

^aThe entropic contribution to the binding free energy is calculated using harmonic approximation ($T\Delta S_{\text{HA}}$) and quasi-harmonic approximation ($T\Delta S_{\text{QHA}}$). All the energy values are listed in kcal·mol⁻¹.

Table 2. Coefficient of Determination (r^2) and MUE between the Experimental and Calculated Binding Energies for Different Radii Sets in MM/PBSA Calculations

	r^2				MUE (kcal·mol ⁻¹)			
	bondi	mbondi	mbondi2	mbondi3	bondi	mbondi	mbondi2	mbondi3
ΔH_{cal}	0.90	0.97	0.90	0.91	11.4	5.5	12.9	9.8
$\Delta H_{\text{cal}} + T\Delta S_{\text{HA}}$	0.77	0.88	0.79	0.78	6.3	12.2	4.9	8.0
$\Delta H_{\text{cal}} + T\Delta S_{\text{QHA}}$	0.90	0.97	0.91	0.92	1.1	6.8	1.2	2.6
$\Delta\Delta H_{\text{cal}}$	0.86	0.96	0.87	0.88	1.4	3.4	1.6	1.4

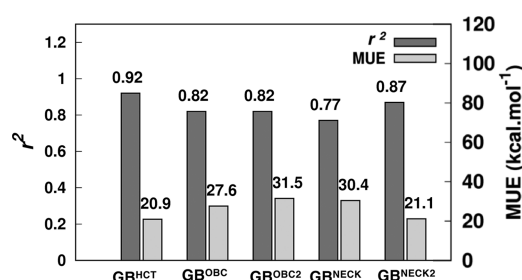
very high in terms of r^2 . However, the MUE between ΔG_{exp} and ΔH_{cal} is fairly low for *mbondi* (5.5 kcal·mol⁻¹) compared to that for *bondi* (11.8 kcal·mol⁻¹), *mbondi2* (12.9 kcal·mol⁻¹), and *mbondi3* (9.4 kcal·mol⁻¹). It is possibly due to the transferability of the *mbondi* radii set from small molecules to large biomolecules, which was tested for the PBSA model only,^{38,39} whereas more advance radii sets (*mbondi2* and *mbondi3*), which performed poorly compared to *mbondi2*, were tested with specific GB models only.^{26,28} From Figure 4, it is

**Figure 4.** Bar graph of correlation (primary y-axis) and MUE (secondary y-axis) between the enthalpy and experimental free energies using different radii sets in MM/PBSA.

apparent that *mbondi* shows the highest correlation ($r^2 = 0.97$) and lowest MUE. Considering both r^2 and MUE as an evaluation criterion, the *mbondi* radii set appears to be a suitable choice for MM/PBSA calculations on protein–glycan complexes.

3.3. Performance of Various GB Models in MM/GBSA.

MM/GBSA shows lower ΔH_{cal} values compared to MM/PBSA. This is due to the electrostatic solvation energy term, which is consistently larger for different GB models than for the PB model. The ΔH_{cal} predictions using the GB^{HCT} model provide the highest r^2 (0.92) and the lowest MUE (20.9) with the ΔG_{exp} than other GB models (Table 3, Figure 5). This finding is in agreement with a recent study³⁷ but differs from a previous one by Wang et al. (2011) where GB^{OBC} yielded a better correlation compared to GB^{HCT} for general ligands.⁴⁰

**Figure 5.** Bar graph of correlation (primary y-axis) and MUE (secondary y-axis) between the enthalpy using different radii sets in the MM/GBSA approach.

The GB^{OBC} and GB^{OBC2} models show a similar r^2 (0.82), which is not surprising as both models have the same computational foundation, and even minor differences in parameters provide comparable results. GB^{NECK2} ($r^2 = 0.87$) performed significantly better than GB^{NECK} ($r^2 = 0.77$), which is expected because GB^{NECK2} includes a correction term for better protein hydrogen bond length distributions and improved solvation energies.²⁸

3.4. Performance of QM–MM/GBSA. The protein–glycan binding is often derived by the CH $\cdots\pi$ stacking interaction between the aromatic amino acid residue and CH from the hydrophobic face of the saccharide.⁴¹ In the RSL system, Trp76 forms a CH $\cdots\pi$ stacking interaction with L-Fuc. In the previous study, it was shown that CH $\cdots\pi$ stacking can contribute up to 5 kcal·mol⁻¹ gain in the binding free energy.⁴² However, because of a lack of any specific term for this particular interaction, the applicability of current protein force fields for protein–glycan binding free-energy calculations is limited. This shortcoming was the motivation to evaluate the performance of symmetrical QM approaches for protein–glycan complexes, mainly density functional theory methods, DFTB, and SCC-DFTB Hamiltonians as they include dispersion effects.

The ΔH_{cal} of monosaccharides using various QM Hamiltonians and GB models is shown in Table S1a–e (see Figure S2a–e and Table S2a–e in the Supporting Information). The r^2 and MUE between ΔG_{exp} and ΔH_{cal} using different GB models and semiempirical approaches, PM3, PM6, MNDO,

Table 3. Coefficient of Determination (r^2) and MUE between the Experimental Binding Energies (ΔG_{exp}) and Calculated Enthalpies (ΔH_{cal}) for Various GB Models in MM/GBSA Calculations^a

method	GB ^{HCT}	GB ^{OBC}	GB ^{OBC2}	GB ^{NECK}	GB ^{NECK2}	GB ^{HCT}	GB ^{OBC}	GB ^{OBC2}	GB ^{NECK}	GB ^{NECK2}
ΔH_{cal}	0.92	0.82	0.82	0.77	0.87	20.9	27.6	31.5	30.4	21.1
$\Delta H_{\text{cal}} + T\Delta S_{\text{HA}}$	0.83	0.77	0.79	0.80	0.83	2.9	9.6	13.5	12.5	3.1
$\Delta H_{\text{cal}} + T\Delta S_{\text{QHA}}$	0.93	0.83	0.83	0.78	0.88	8.6	15.2	19.2	18.1	8.8
$\Delta\Delta H_{\text{cal}}$	0.89	0.76	0.76	0.69	0.84	2.8	1.4	1.8	2.7	1.9

^a ΔG_{cal} ($\Delta H_{\text{cal}} + T\Delta S$) is compared with ΔG_{exp} and $\Delta\Delta G_{\text{cal}}$ has been compared with $\Delta\Delta H_{\text{cal}}$.

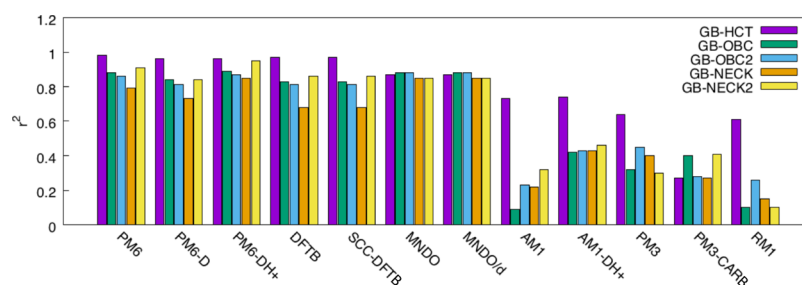


Figure 6. Bar graph of the correlation between the calculated enthalpy and experimental binding free energies for every combination of QM Hamiltonian with five GB models in the QM–MM/GBSA calculations.

Table 4. Coefficient of Determination (r^2) and MUE between the Experimental Binding Free Energy and Calculated Enthalpy (ΔH_{cal}) for Various QM Models Combined with Different GB Models in QM–MM/GBSA Calculations

QM models	r^2					MUE (kcal·mol ^{−1})				
	GB ^{HCT}	GB ^{OBC}	GB ^{OBC2}	GB ^{NECK}	GB ^{NECK2}	GB ^{HCT}	GB ^{OBC}	GB ^{OBC2}	GB ^{NECK}	GB ^{NECK2}
PM6	0.98	0.88	0.86	0.79	0.91	4.7	1.6	3.4	4.4	3.5
PM6-D	0.96	0.84	0.81	0.73	0.84	7.3	10.5	15.8	16.5	8.7
PM6-DH+	0.96	0.89	0.87	0.85	0.95	17.7	20.8	26.1	26.6	19.0
SCC-DFTB	0.97	0.83	0.81	0.68	0.86	2.5	2.3	6.0	7.3	3.6
DFTB	0.97	0.83	0.81	0.68	0.86	2.5	2.3	6.0	7.3	3.6
MNDO	0.87	0.88	0.88	0.85	0.85	83.0	82.5	80.1	81.2	89.1
MNDO/d	0.87	0.88	0.88	0.85	0.85	83.0	82.5	80.1	81.2	89.1
AM1	0.73	0.09	0.23	0.22	0.32	7.0	6.5	3.2	4.5	8.3
AM1-DH+	0.74	0.42	0.43	0.43	0.46	21.7	22.4	25.8	24.5	20.7
PM3	0.64	0.32	0.45	0.40	0.30	1.7	1.6	3.0	2.0	3.3
PM3-CARB	0.27	0.40	0.28	0.27	0.41	48.6	45.0	38.9	38.9	50.9
RM1	0.61	0.10	0.26	0.15	0.10	7.1	6.1	3.0	4.0	9.8

DFTB, AM1, and RM1, are shown in Figure 6 and Table 4. PM6, DFTB, and SCC-DFTB in combination with GB^{HCT} show a passable improvement in r^2 and MUE when compared to the best performing combination of MM/PBSA (*mbondi* radii) and MM/GBSA (GB^{HCT} model). Other QM Hamiltonians resulted in the poor r^2 and MUE compared to MM/PBSA and MM/GBSA. This shows that the energies of the QM zone have a large impact on the overall performance of QM–MM/GBSA, and only a suitable QM method can provide meaningful predictions.

3.5. Effect of QM Theory on QM–MM/GBSA. The ΔH_{cal} calculated using the semiempirical approach PM6 and DFTB in combination with the model GB^{HCT} provides the most acceptable agreement with experimental binding energies (Figure 6, Table 4). As expected, PM6 outperformed PM3 and AM1. This is likely as PM6 Hamiltonian encompasses multiple corrections in PM3 and AM1.^{33,43} In contrast to the study on benzimidazole FabI inhibitors, where PM3 performs much better than PM6, RSL–glycan binding is represented well by PM6.³⁷ The deficient performance of PM6 in Su et al. was assumed to be due to the parameters of PM6, which are adopted from PM3 in Amber. However, an adequate performance of PM6 in RSL–glycan complexes suggests that the magnitude of the parameter α adopted from PM3 may not be critical for the accuracy of PM6. Further studies using PM6 are therefore required to check the validity of the coefficient α used in PM6.

The MNDO and MNDO/d provide a qualitatively good agreement with experimental binding energies ($r^2 = 0.87$) and show minimal effect of the GB model it was used with. However, the absolute binding free energies used have very high positive numbers, suggesting all the ligands to be

technically nonbinders (Figure S2a–e in the Supporting Information). Obtaining binding energies that are off from the experimental range is always fallacious for even qualitative analysis of ligand binding. As expected, there is no gain in the performance of the more recent MNDO/d methods, which add d-orbitals for the description of transition metals.³⁰

Interestingly, AM1 with GB^{HCT} performed slightly better than PM3 that is known to handle hydrogen bonds more accurately than AM1. A possible explanation for this may be that the binding strength in RSL–glycan complexes is prevailed by the CH $\cdots\pi$ stacking interaction and hydrogen binding plays only a subsidiary role.⁴² PM3 performed slightly better when used with the GB model than GB^{HCT}. The overall response of PM3-CARB, a method that was developed to accurately model carbohydrate analogues, was unexpectedly worse than PM3. It is possible that inaccuracies are due to the amino acid residues in the QM zone. Because the main idea behind using QM–MM Hamiltonian was to check if the CH $\cdots\pi$ interaction between Trp76 and the monosaccharide is captured well, we excluded the possibility of having a monosaccharide in the QM zone in this case. Further calculations on a different protein are needed to determine exactly why the performance of PM3-CARB is affected because of the inclusion of amino acid residues in the QM zone.

3.6. Effect of Entropy in Calculations. We calculated entropies using quasi-harmonic approximation with ptraj ($T\Delta S_{\text{QHA}}$) and harmonic approximation with nmode ($T\Delta S_{\text{HA}}$). The MM/PBSA and MM/GBSA results show that the ΔG_{cal} obtained using ΔS_{HA} weakens the agreement with ΔG_{exp} compared to ΔS_{QHA} in all the calculations (Tables 2 and 3). The $T\Delta S$ from harmonic approximation is nearly a constant value of ~ 12.3 kcal·mol^{−1} for all ligands (Table 1).

Table 5. Coefficient of Determination (r^2) and MUE between Relative Change in Experimental Binding Free Energy ($\Delta\Delta G_{\text{exp}}$) and Relative Change in Calculated Enthalpy ($\Delta\Delta H_{\text{cal}}$) for Various QM Models Combined with Different GB Models in QM–MM/GBSA Calculations

QM models	r^2					MUE (kcal·mol ^{−1})				
	GB ^{HCT}	GB ^{OBC}	GB ^{OBC2}	GB ^{NECK}	GB ^{NECK2}	GB ^{HCT}	GB ^{OBC}	GB ^{OBC2}	GB ^{NECK}	GB ^{NECK2}
PM6	0.98	0.85	0.83	0.72	0.91	1.32	0.81	0.68	0.92	1.09
PM6-D	0.94	0.79	0.76	0.64	0.79	2.44	0.91	1.32	1.76	2.00
PM6-DH+	0.94	0.87	0.85	0.80	0.94	3.46	1.92	2.35	2.99	3.16
SCC-DFTB	0.97	0.77	0.75	0.57	0.82	0.38	1.18	1.03	1.14	0.97
DFTB	0.97	0.77	0.75	0.57	0.82	0.38	1.18	1.03	1.14	0.97
MNDO	0.85	0.86	0.86	0.84	0.84	10.80	11.34	11.11	10.38	11.23
MNDO/d	0.85	0.86	0.86	0.84	0.84	10.80	11.34	11.11	10.38	11.23
AM1	0.82	0.24	0.37	0.23	0.32	2.71	3.82	3.58	2.92	2.88
AM1-DH+	0.69	0.38	0.39	0.35	0.35	1.08	1.97	1.86	1.72	1.53
PM3	0.53	0.11	0.27	0.19	0.08	0.97	1.55	1.36	1.44	1.44
PM3-CARB	0.47	0.51	0.42	0.52	0.57	3.62	4.73	4.11	3.73	4.51
RM1	0.48	0.01	0.13	0.02	0.00	1.49	2.29	2.14	1.86	1.90

This is in good agreement with that of Su et al., where inclusion of entropy (ΔS_{HA}) compromises the agreement between the predicted absolute binding free energy and the calculated binding free energy.³⁷ This may be due to the large fluctuations that are often seen in the entropic calculations. However, monosaccharides have a well-defined conformation, so our results are less affected by this frailty in the entropy calculations. The inclusion of the ΔS_{QHA} entropy has very marginal improvement in r^2 between the calculated and experimental energies (Tables 1, 2, and S6). The MUEs are also marginally improved in some cases. The entropy calculations typically dominate the computational cost of the binding free-energy calculations, and this is why this term has been repeatedly omitted in the MM/PBSA and MM/GBSA calculations.¹⁶ It is also discussed widely that the nature of MM/PBSA and MM/GBSA calculations is better suited for ranking the ligands in their order of binding than obtaining their absolute free energy. These results also suggest that it is cogent to neglect the entropic term while aiming to achieve a ranking order of the glycan binding. The inclusion of entropic terms may be reasonable only when aiming to obtain a quantitatively better agreement with experimental values, that is, lower MUE with experimental values.

3.7. Relative Binding Free-Energy Calculations. It has been widely proclaimed that the ΔG_{cal} by MM/GBSA or MM/PBSA may not be always in agreement with ΔG_{exp} , but their ability to rank the ligands based on the calculated $\Delta\Delta G_{\text{exp}}$ often agrees well with their experimental values.^{6,16} This motivated us to calculate the change in the binding free enthalpies of monosaccharides relative to MeFuc ($\Delta\Delta H_{\text{cal}}$). This is similar to the study of Mishra et al. (2015), where it was shown that TI can predict $\Delta\Delta G_{\text{cal}}$ with accuracy greatly improved compared to LIE and several other docking approaches.¹² Comparison of $\Delta\Delta H_{\text{cal}}$ with $\Delta\Delta G_{\text{exp}}$ shows a negligible difference in terms of r^2 , but MUEs are much smaller for $\Delta\Delta H_{\text{cal}}$ than for ΔH_{cal} or ΔG_{cal} (Tables 2, 3, 5, and S4). A careful comparison of the minor difference in r^2 for $\Delta\Delta H_{\text{cal}}$ shows that in general, it is either very similar or marginally lower than r^2 for ΔH_{cal} . This trend is rather similar for MM/PBSA, MM-GBSA, and QM–MM/GBSA calculations, suggesting a comparable performance of these methods for both ΔH_{cal} and $\Delta\Delta H_{\text{cal}}$ calculations.

4. CONCLUSIONS

In this study, we have examined the binding free-energy prediction capabilities of the MM/PBSA, MM/GBSA, and QM–MM/GBSA approaches on protein–glycan complexes. We have carefully investigated the impact of MD simulation length, van der Waals radii set, GBSA models, QM Hamiltonians, and entropic contributions on the estimation of absolute and relative binding free energies.

The MM/PBSA and MM/GBSA calculation on 100 frames taken from a minimum of 7 ns MD simulation provided a statistically good agreement with the experimental binding free energies. MM/PBSA using *mbondi* radii sets offers the best correlation ($r^2 = 0.96$) with the experimental values, but the effect of radii sets on the predictive power of MM/PBSA is lenient ($r^2 = 0.89$ for *bondi* and *mbondi2*). The GB^{OBC} model in MM/GBSA shows the highest correlation ($r^2 = 0.91$), but the performance is relatively weaker than MM/PBSA. The QM–MM/GBSA using the GB^{HCT} model and PM6 or DFTB Hamiltonian resulted in a marginally improved agreement ($r^2 = 0.96$) with the experimental binding energies compared to MM/PBSA with the *mbondi* radii set. The QM Hamiltonians have a notable impact on the QM/MM-GBSA estimates (Table S1a–e). Because the DFTB/SCC-DFTB calculations are computationally much more demanding than PM6, it may be worth using PM6 for virtual screening involving tasks where thousands of compounds need to be calculated.

The inclusion of entropic terms from the quasi-harmonic approximation (ΔS_{QHA}) provides no meaningful improvement in the agreement between the predicted and experimental binding free energies, whereas the harmonic approximation (ΔS_{HA}) entropies compromised the agreement with the experimental binding free energies. It is important to discuss here that the inclusion of a larger number of frames in ΔS_{HA} calculation might increase the agreement with the experimental free energies, but considering the computational cost of entropic calculation, we could not afford to test it under current circumstances. The purpose of such a calculation is to use them for ranking of the compounds; thus, the entropic contributions can be safely ignored as the ΔH calculation alone performs quite well in ranking the binding order of the ligands.

In brief, the performance of implicit solvent methods on the protein–glycan complex was found to be highly dependent on the choice of the parameters, implicit solvent models, and the MD simulation length. Our results suggest that MM/PBSA

(*mbondii* radii), MM/GBSA (GB^{HCT} model), and QM–MM/GBSA (GB^{HCT} and PM6/DFTB Hamiltonian) calculations with entropic terms can be an appropriate choice for binding free-energy calculations in protein–glycan complexes. Because the CH $\cdots\pi$ stacking is frequently observed in glycan binding, we suggest QM–MM/GBSA using DFTB Hamiltonian to be used to study such protein–glycan complexes. However, PM6 is computationally more feasible for the tasks such as for virtual screening of the large databases. It may provide a better computational cost to performance ratio. We do not observe any significant difference in the prediction ability of these methods for ΔG_{cal} or $\Delta\Delta G_{\text{cal}}$. However, MUE is significantly lower while calculating $\Delta\Delta G_{\text{cal}}$. While comparing the $\Delta\Delta G_{\text{cal}}$ prediction ability of end-point methods (MM/PBSA, MM/GBSA, and QM–MM/GBSA) with the alchemical approach (TI), a slightly better correlation is noted, but the MUEs are much higher compared to TI. Although we needed only good sampling and good force field in TI, the MM/GBSA type of calculations also require an optimized workflow to make a reliable prediction. Thus, if the experimental binding free-energy data are available, it is recommended to optimize and find a computational workflow for particular systems rather than using all the default settings available in the software. Because this group of bacterial lectins (RSL, AFL, and AAL) is an excellent drug target against bacterial infections,^{9,44} the lessons and workflow from this study can be used further in designing new antiadhesion compounds.

■ ASSOCIATED CONTENT

■ Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jpcb.8b03655.

ΔH of MeFuc binding, ΔH of binding free energies by different QM models, bar graphs of correlation (primary y-axis) and MUE using different QM–MM Hamiltonians and different QM models, plot of correlation between the experimental and calculated enthalpy for various QM models, coefficients of determination (r^2) and MUE, plot of correlation between change in experimental binding energy and change in calculated enthalpy for various QM models, and plot of correlation between the experimental binding free energy and the calculated free energy for various QM models (PDF)

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Author Contributions

S.K.M. and J.K. conceived and designed the calculations. S.K.M. performed the calculations, analyzed the data, and wrote the manuscript. Both authors have given approval to the final version of the manuscript.

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Notes

The authors declare no competing financial interest.

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